

Run 12 | NME

India

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0

Sample 27,555 respondents **Coverage** 1 country; 2015–2022

Positive cases 1,849 (6.71%) **Case time** 1m 01s

Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9878). Highest F1: XGBoost (0.7678). For XGBoost, adding controls changed ROC-AUC by +0.0083 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9878	0.8258	96.86	76.06	77.51	0.7678	<1s
CatBoost	0.9878	0.8288	96.82	75.80	77.24	0.7651	13s
H2O GBM	0.9877	0.8227	96.46	69.86	82.93	0.7584	3s
Random Forest	0.9873	0.8177	96.41	69.13	83.74	0.7574	<1s
Lasso	0.9866	0.8053	96.59	73.26	77.24	0.7520	12s
Elastic Net	0.9865	0.8049	96.62	73.52	77.51	0.7546	13s
Logistic Regression	0.9863	0.8029	96.66	73.90	77.51	0.7566	1s
Ridge	0.9858	0.8052	96.66	73.78	77.78	0.7573	11s
Decision Tree	0.9857	0.7648	96.68	76.27	73.17	0.7469	<1s
Naive Bayes	0.9039	0.4289	93.23	49.56	60.70	0.5457	5s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	668.64	41.01
2	Established business owner	262.75	16.12
3	Occupational status	73.07	4.48
4	Age	68.66	4.21
5	Survey year	51.74	3.17
6	Household size	43.31	2.66
7	Population	37.41	2.29
8	GDP per capita (PPP)	36.77	2.26

Share is the percentage of total forest impurity importance across all 40 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 12_obj1_india_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.